

Uber Request Data Business Insights Report

1. Executive Summary

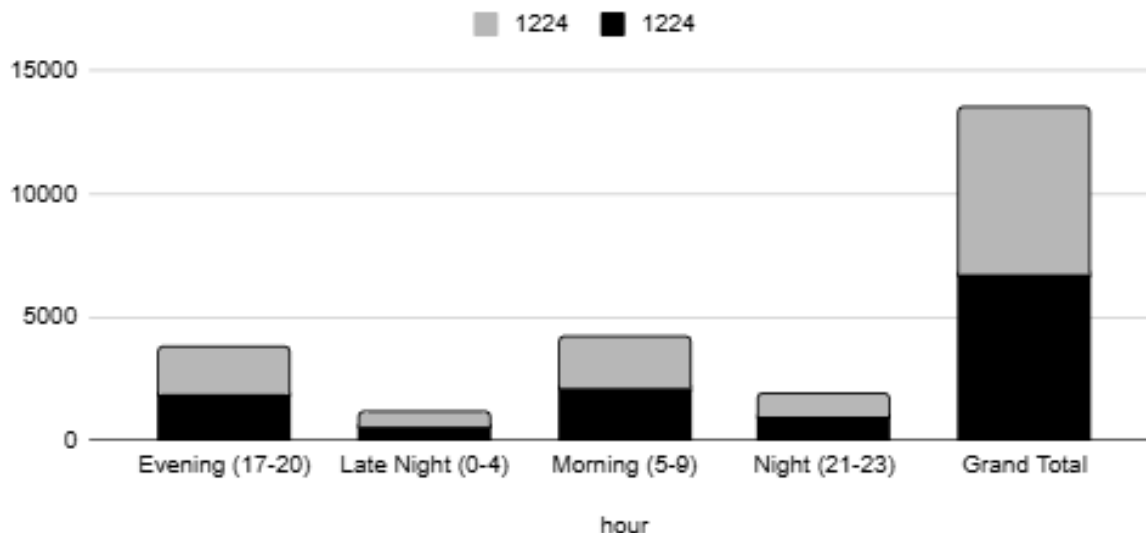
This report analyses 6,745 Uber ride requests recorded between July 11–15, 2016, across two pickup locations the City and the Airport. The core finding is a severe and persistent supply-demand mismatch: only 42% of all requests resulted in a completed trip, meaning more than half of all demand went unserved across the five-day period. The problem manifests differently by location and time of day. Airport pickups suffer from an acute shortage of available cars, particularly during evening hours (5–9 PM). City pickups face a high driver cancellation rate, most concentrated during the morning rush (5–9 AM). Afternoon hours (10 AM–4 PM) represent the only healthy window, with completion rates reaching 59–67%.

Key Finding Value

- Implication Overall Completion Rate 42% Majority of demand is lost
- No Cars Available Rate 39% Primary supply failure mode
- Driver Cancellation Rate 19% Secondary failure — City morning peak
- Worst Time Slot Evening (17–20h) 66% of requests go unserved
- Best Time Slot Afternoon (10–16h) 59% completion — adequate supply
- Airport Unserved Rate ~53% No-car problem dominates airport
- City Cancellation Rate ~30% Driver behaviour issue in City

Fig 1.1 — Hourly Request Distribution by Status:-morning rush (5–9 AM) and evening surge (17–21h). The evening surge has the highest volume of all time windows but is dominated by "No Cars Available", confirming the supply crisis.

Hourly Request by Status



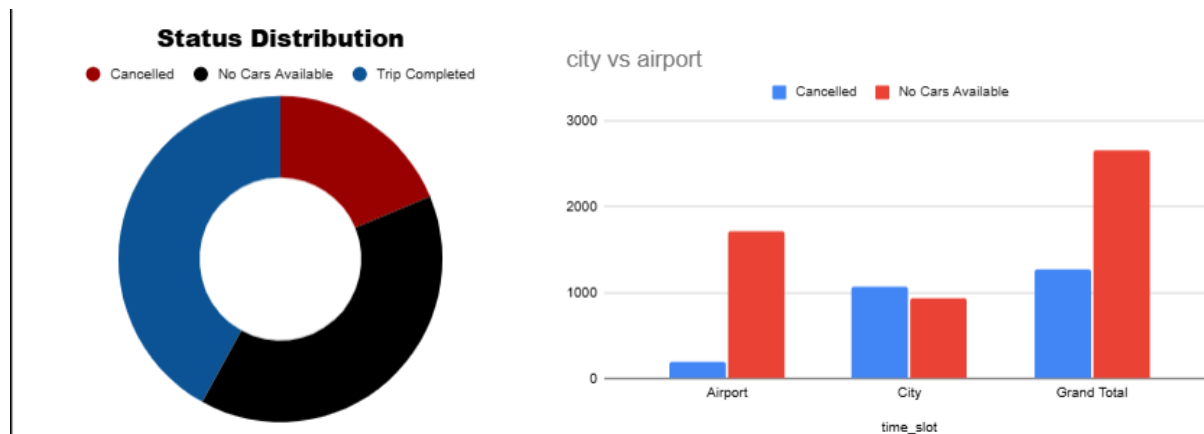
2. Business Problem The fundamental business problem Uber faces in this dataset is a structural imbalance between rider demand and driver supply. While total demand is strong and consistent across all five days (averaging ~1,350 requests per day), supply fails to meet it at predictable, recurring time windows. This is not a random failure — it is a systematic pattern that repeats daily.

2.1 Two Distinct Failure Modes The data reveals two separate problems operating simultaneously, each concentrated in a different location and time window:

Failure Mode 1 — No Cars Available (Airport, Evening): During the evening surge (17–21h), the Airport experiences a near-complete absence of available drivers. Of the 1,127 "No Cars Available" records in the Evening slot, the majority originate from Airport pickups. Passengers arriving on flights cannot get rides home.

Failure Mode 2 — Driver Cancellations (City, Morning): During the morning rush (5–9 AM), City pickup cancellations spike dramatically — 843 cancellations in the Morning slot alone, accounting for 67% of all cancellations in the dataset. Drivers are accepting requests but then abandoning them.

2.1



2.2 Supply Gap by Time Slot

Slot	Total Requests	Trip Completed	Cancelled	No Cars Availab	Unservd	Unservd %	Completion %	Supply Gap	sanity
2	Late Night (0-4)	578	214	65	299	364	63	37	Moderate
3	Morning (5-9)	2103	854	843	406	1249	59.4	40.6	Critical
4	Afternoon (10-16)	1224	722	168	334	502	41	59	Critical
5	Evening (17-20)	1893	642	124	1127	1251	66.1	33.9	Critical
6	Night (21-23)	947	399	64	484	548	57.9	42.1	Critical

3. Key Business Insights

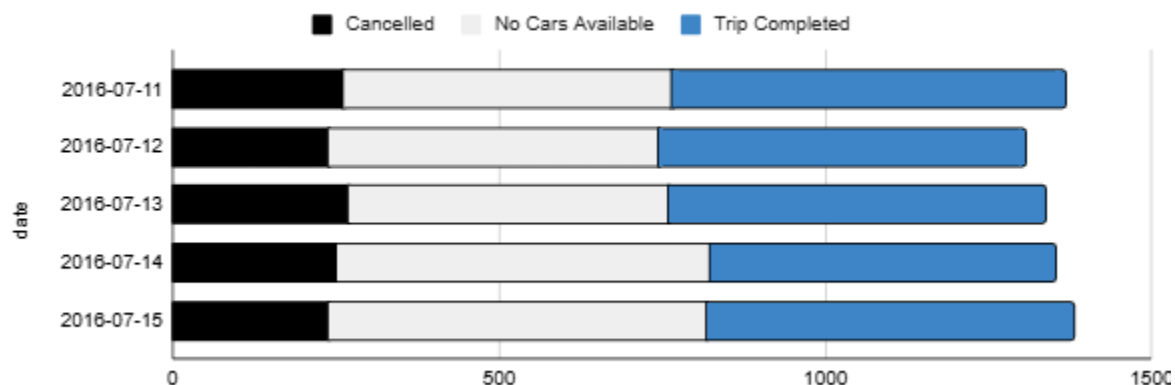
Insight 1 Evening Airport Crisis is the #1 Revenue Loss Window The Evening slot (17–20h) at the Airport is where the most revenue is being lost. 66% of evening requests go unserved, and the dominant reason is no car availability — not cancellations. This means passengers are ready and willing to pay, but Uber simply has no drivers positioned at the Airport during peak arrival hours. This is a pure supply positioning failure.

Insight 2 Driver Cancellations Are Destroying City Morning Performance The City Morning slot records 843 cancellations — more than all other slots and locations combined. Drivers are accepting City requests during the morning rush and then cancelling them. This suggests drivers may prefer Airport trips (longer, higher fare) and cancel City short-trips. This behaviour actively harms the completion rate and damages rider trust.

Insight 3 Afternoon is the Only Healthy Window — A Benchmark to Protect Afternoon (10–16h) achieves a 59% completion rate — the best of all slots. Supply and demand are relatively balanced during this window. This should serve as the baseline target. Understanding what makes Afternoon work (driver availability, lower cancellations, balanced City/Airport mix) can inform strategies for fixing the problem slots.

Insight 4 Performance Degrades Toward End of Week Jul 11–13 all achieve completion rates above the 42% average (44%, 43%, 43.2%). Jul 14 and 15 drop below average (39.2%, 40.6%). This suggests driver fatigue, reduced availability, or demand shifts in the latter half of the week. A 5-day dataset limits conclusions, but the trend warrants monitoring over a longer period.

Daily Trend and Status



4. Daily Performance Scorecard Daily request volumes remained consistent across the five-day period, averaging 1,349 requests per day. However completion rates tell a declining story — the first three days exceeded the 42% average while the final two fell below it.

Date	Total Requests	Completed	Unservd	Completion %	vs Avg (42%)	Performance
2	2016-07-11	1367	601	766	44	2 Above Average
3	2016-07-12	1307	562	745	43	1 Above Average
4	2016-07-13	1337	577	760	43.2	1.2 Above Average
5	2016-07-14	1353	530	823	39.2	-2.8 Below Average
6	2016-07-15	1381	561	820	40.6	-1.4 Below Average

5. Recommendations Based on the analysis, the following targeted interventions are recommended. Each addresses a specific, data-identified failure point rather than applying blanket supply increases.

1 Airport Evening Surge Pricing & Driver Incentives Deploy surge pricing at the Airport between 17:00–21:00 to attract more drivers to position themselves at the Airport ahead of the peak. Offer guaranteed bonuses for Airport pickups during this window. Expected impact: reduce the 66% unserved rate in the Evening slot by drawing drivers from lower-demand areas

2 Anti-Cancellation Policy for City Morning Requests Introduce a penalty or monitoring flag for City cancellations during 05:00–10:00. With 843 cancellations in this single window, a targeted policy here would have the highest per-intervention impact of any recommendation. Consider mandatory acceptance rates during morning peak hours for City-based drivers.

3 Night Shift Driver Programme The Night slot (21–23h) has a 57.9% unserved rate driven almost entirely by no-car availability. Introducing a dedicated night-shift pool of drivers with premium night rates would directly address this. Uber can pilot this with a small cohort of willing drivers and measure completion rate improvement.

4 Demand Forecasting and Proactive Positioning The demand peaks at 5–9 AM and 17–21h are perfectly predictable. Uber should use this data to pre-position drivers near the Airport before 17:00 and in the City before 05:00, rather than reacting to demand after it appears. Automated driver nudge notifications 30 minutes before peak windows can help achieve this.

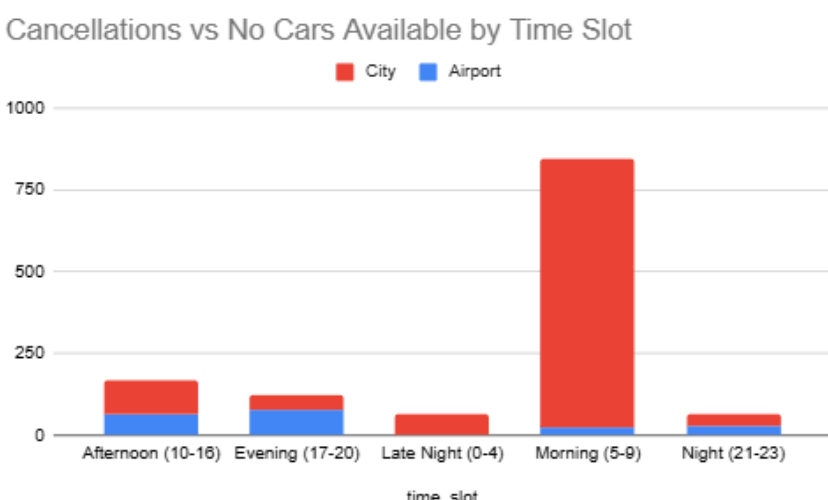


Fig 5 — Cancellation & No Cars Available by Time Slot